

Finite element crack propagation calculation using trial cracks

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Abstract

We present a method to perform finite element calculations for crack propagation problems with arbitrary crack directions in two dimensions. The crack direction (angle of propagation) is determined by inserting small “trial cracks” at the crack tip and measuring the energy release rate when these trial cracks are opened. Several finite element calculations are performed for each crack length increment. In this way, the energy release rate can be calculated even in cases where other methods fail. After explaining the algorithm, it is applied to two test problems: a three-point bending specimen with central or offset initial crack and crack propagation near an elastic inclusion.

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1. Introduction

The finite element method is frequently used to study problems of fracture and crack propagation. In performing a finite element simulation of a propagating crack, two issues have to be addressed:

- implementing the crack propagation in the FE code,
- determining the crack propagation direction.

The first problem can be solved by remeshing the geometry so that the propagating crack travels directly on the bounding line between elements. The crack propagation itself is then performed by releasing the nodes between the elements (debonding). This is the option chosen here. The other possibility is to use enhanced elements that are able to directly represent a discontinuity inside the elements (partition-of unity or XFEM methods [1,2]). Although these methods are very powerful, they are not yet imple-

mented in standard FEM codes, and a considerable effort is therefore needed to use them, especially in situations with large deformations and non-linear material properties.

A more physical problem is to actually choose the direction in which the crack is going to propagate. Different criteria have been used in the past, summarised in [3]. The maximum circumferential stress criterion (trying to take into account K_I and K_{II}) is a local approach. Other criteria are based on the assumption that a crack proceeds in the direction which maximises the (global) energy release rate. The main disadvantage of these criteria is that the energy release rate cannot be calculated directly from the stress field. To circumvent this problem, strategies like the virtual crack extension method [4] have been invented.

One especially simple method to calculate the energy release rate is to actually perform a test simulation with real crack extension and to calculate the energy release rate from this. The disadvantage of this is that two calculations are needed for each crack length increase and that a strong mesh refinement at the crack tip is also required.

The approach presented in this paper is based on this idea. Instead of performing just one crack extension calculation to get at the energy release rate, we perform several

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such calculations with the crack propagating in different directions. The energy release rates for the different propagation directions are compared, and the optimum direction is chosen as actual crack direction. Obviously, this method suffers from the disadvantage of strongly increasing computing times (due to the required mesh refinement at the crack tip and the need to perform several calculations), but it may nevertheless be worthwhile in situations where other methods are difficult to use, for example in heterogeneous materials. Examples for such situations will be discussed in the Section 4.

2. Crack propagation algorithm

2.1. Trial crack propagation

To simplify the discussion, we assume that no changes in the external load occur during crack propagation and that the problem is time-independent, i.e., no visco-elastic or visco-plastic properties or inertial forces are involved. (These assumptions are not required for the method to work, see the Section 4 for a further discussion.)

The algorithm proceeds as follows: Assume that an initial crack is present and the direction of crack propagation is to be calculated, using the maximum energy release rate. In two dimensions, this is a one-dimensional optimisation problem, where the optimisation variable is the angle describing the propagating crack and the energy release rate is a function of this angle.

Several calculations, each propagating one trial crack by a fixed distance δa , are performed, and the energy release rate is calculated for each. The crack direction is varied until a local maximum of the energy release rate has been found with sufficient precision. The optimum crack direction is then chosen for the next crack increment. (Some care has to be taken to ensure that all cracks have the same absolute length. It is best to calculate the actual crack length and to normalise by it to account for slight irregularities in the crack length due to different meshes.)

2.2. Optimisation problem

To find the optimum crack direction, a Brent algorithm, described in detail in [5], is used. This algorithm starts with a local maximum bracketed between two other values and refines the maximum position by decreasing the size of the bracketing interval.

We start by performing three trial crack calculations at angles of 0° and $\pm 5^\circ$ to the current crack direction (see Fig. 1). If the energy release rate of the medium value (0°) is not larger than that of the other two values, the maximum is not yet bracketed. In this case, another trial crack configuration (at larger angle relative to 0°) is calculated on the side where the release rate is maximum. This is repeated until a maximum is bracketed.

We now have three values for the crack energy. A parabolic interpolation of the three data points is performed to

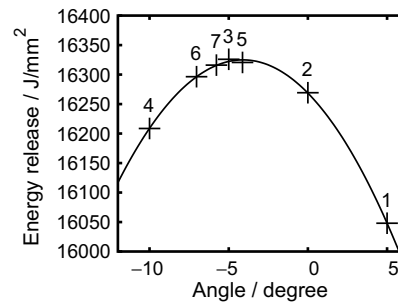


Fig. 1. Bracketing of the maximum of the energy release function using the Brent algorithm. The maximum is bracketed in steps 1–4, afterwards the boundaries are drawn closer together. The final value chosen was number three. The line shows a parabolic fit through the data points.

calculate a new estimate of the maximum. If this is successful (the calculated point is inside the boundaries and not too close to a value already calculated), a trial crack propagation is performed at this angle. Otherwise, the new angle is calculated using a golden section search. Calculation stops if maximum is bracketed with a precision of $\pm 0.5^\circ$. Details on the algorithm can be found in [5].

The algorithm used here deviates in one respect from that described in [5]. There, the first step after the maximum which is bracketed is always a golden section search. Here, instead, we start by trying a parabolic interpolation first to exploit information about crack energy already obtained.

In the examples shown below, 4–9 trial crack calculations were always sufficient to calculate the crack direction with the desired accuracy.

2.3. Remeshing

Because the direction of the trial cracks is arbitrary, a remeshing of the geometry is needed for each trial crack calculation to ensure that the crack can travel exactly on the nodes between elements. A C++ preprocessor for ABAQUS [6] has been written that can handle the ensuing problems of meshing regions consisting of different materials with a mesh refinement at the crack tip. The algorithm works well even when the mesh refinement zone in front of the crack tip overlaps with the boundary of different material regions. Cracks and debonding surfaces can also be included automatically. Interpolation from the old to the new mesh is performed automatically by ABAQUS after restarting the calculation. The preprocessor is described in more detail in [7].

3. Results

3.1. Three-point bending without offset

We start by checking whether the precision achieved by the algorithm is sufficient in a case where the crack path can be predicted exactly. The test problem is a three-point bending specimen loaded centrally and supported at its

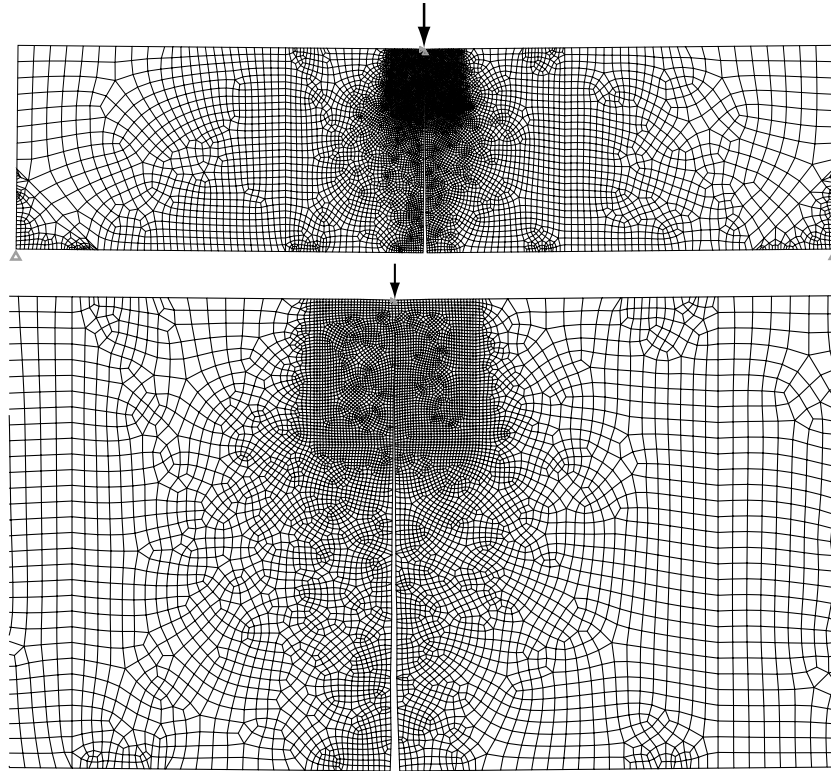


Fig. 2. Calculated crack path for a three-point bending specimen with central initial crack. The deviation from a straight line is small.

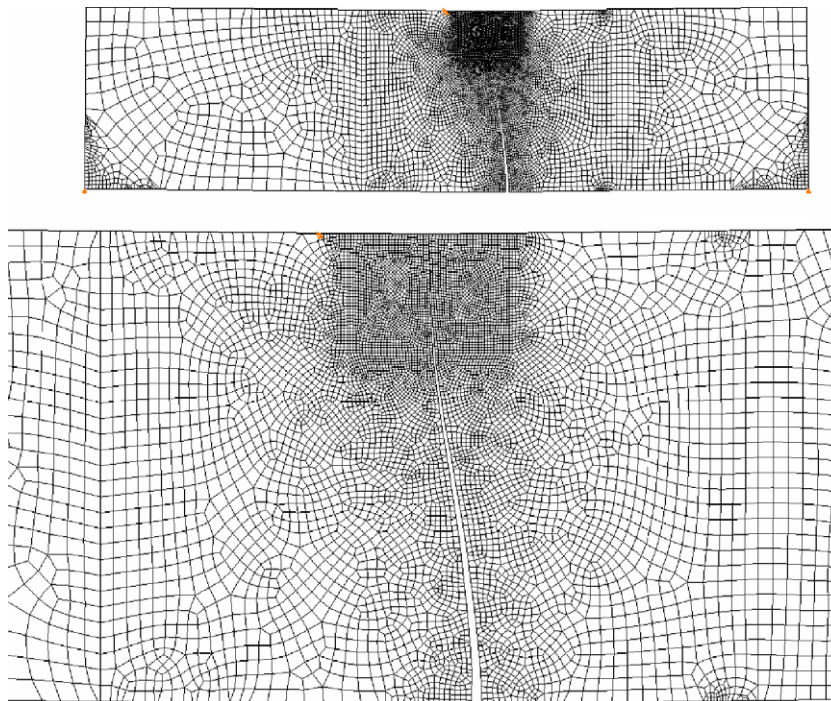


Fig. 3. Calculated crack path for a three-point bending specimen with offset initial crack.

corners. The width of the specimen is 600 mm, the height is 150 mm. (The geometry was taken from [8].)

An initial crack was introduced directly below the loading point so that the crack path should be exactly straight.

Loading was done by initially displacing the loading point by 3 mm and fixing it at this position. The trial crack path length was chosen as 10 mm and the precision of the Brent algorithm as $\pm 0.5^\circ$.

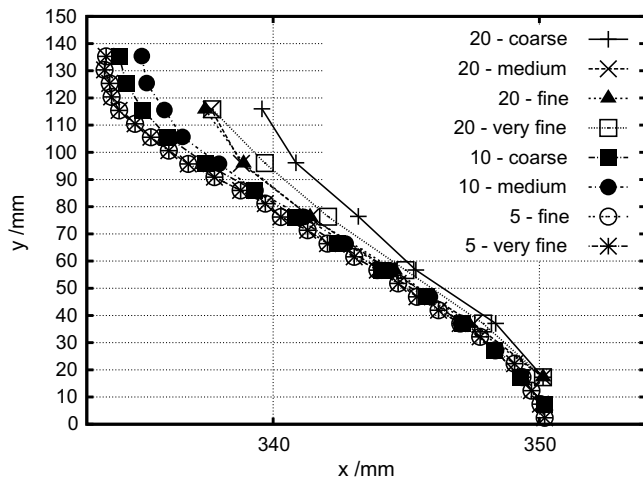


Fig. 4. Calculated crack paths for a three-point bending specimen with offset initial crack, depending on the trial crack path length (5, 10, or 20 mm) and the mesh density. Note that the curvature of the crack path is strongly exaggerated to make differences in the results visible.

Fig. 2 shows the resulting crack path which follows a straight line rather precisely. The largest horizontal deviation from the straight line was by slightly less than 0.5 mm, the maximum angular deviation observed was 2.2° , which is a bit larger than the precision of the Brent algorithm. This shows that in principle the crack path is found rather accurately, but obviously some slight imprecision is present, probably because of an insufficient mesh density.

3.2. Three-point bending with offset

Next, the same problem is studied with a crack initially offset by 50 mm to the right. For this configuration, a plot of the energy release versus the crack angle for the first step was shown above in Fig. 1. The energy is in fact approximated quite well by a parabola, as expected also from [3].

Fig. 3 shows the crack path followed for this configuration which looks plausible. As the path is now curved, it has to be expected that it depends on the trial crack length δa . Fig. 4 shows the dependence of the trial crack on this and on the mesh density. It can be seen that the dependence on the trial crack length is strong if the trial crack length is large.¹ The difference between trial crack length 10 mm and 5 mm is rather small, suggesting that the crack path has (almost) converged. There is also some dependence on the mesh density for larger trial crack lengths (very pronounced for trial crack length 20 mm, less so for 10 mm), but the curves for $\delta a = 5$ mm shows no mesh dependence.

In summary, it can be stated that the algorithm seems to work quite well for this example, provided that the trial crack length is chosen sufficiently small.

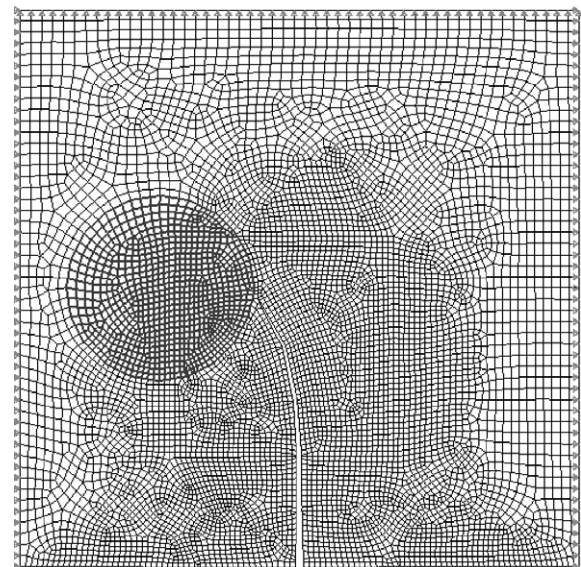
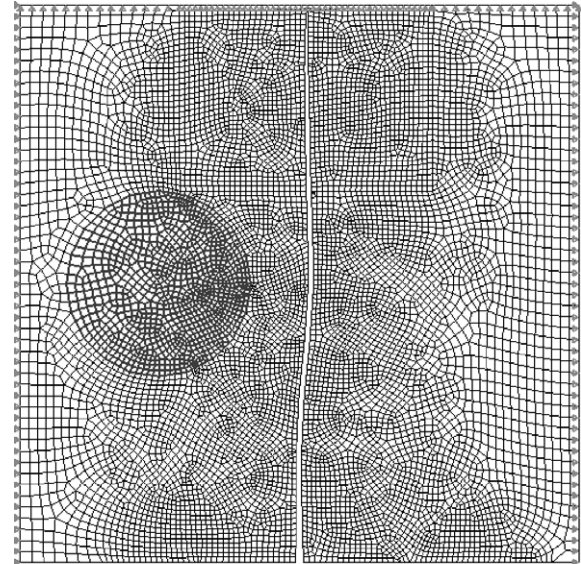


Fig. 5. Calculated crack path for a plate containing a hard or soft inclusion. Trial crack length was 1/10th of the plate edge length. A hard inclusion is avoided by the crack, a soft inclusion is approached.

3.3. Particle inclusion

As further examples, we consider the case of a quadratic plate (edge length 100 mm) containing a circular inclusion and an initial crack. The Young's modulus of the inclusion was chosen either as ten times larger or ten times smaller than that of the matrix material. The plate was fixed vertically at its upper end and displaced horizontally on both sides by 0.5 mm. Fig. 5 shows the resulting crack path for both situations. As expected, the crack is repelled from the particle with large modulus and attracted by the compliant particle. The results are in qualitative agreement with those shown in [3] for a similar problem. So far, penetration of the inclusion by the crack is not implemented in the preprocessor.

¹ Note, however, that the scale in the figure is not homogeneous. If a homogeneous scale is chosen for both axes, the paths are almost indistinguishable.

4. Conclusions and outlook

In this paper, a method to calculate crack propagation problems by using trial cracks has been presented. The method seems to work well in the test cases studied, although the dependence on the trial crack length has to be kept in mind.

The main advantage of the method is that it is applicable to mixed-mode problems and to problems involving several or heterogeneous materials. For the latter case, the preprocessor has to be improved to allow material separation exactly at the interface between different materials.

Currently, the method is implemented using remeshing of the structure for each trial crack. However, it could also be used together with an XFEM method [2,8] where the cracks can travel in arbitrary directions without remeshing. Some remeshing might still be required to ensure a sufficient mesh density near the crack tip even in this case, however.

In the examples studied here, it was assumed that the material was brittle and that the crack propagates regardless of any unloading occurring during its growth. If this is not the case, the crack has to be stopped if the calculated energy release rate is smaller than the critical value of the material considered. The load has then to be increased incrementally after the crack has stopped at the current load, and a new trial crack calculation has to be performed for the new load value. An alternative is to perform one calculation with increased load but without any crack propagation and to calculate the energy release rate by comparing the situation with and without trial cracks but with changes in the load. This latter approach could also be used to calculate energy release rates in cases where the material is globally unloaded, something which is not possible with many other approaches (for example, J -integral methods).

Despite its advantages, the method has three main problems:

- The CPU time is larger by a factor of about six for the cases studied here, compared to a calculation using another crack direction criterion. The CPU time might slightly increase further if changes in the loads occur during crack propagation (see above).
- If material parameters are time-dependent, the crack propagation speed has to be taken into account as well as the direction, resulting in a more involved two-dimensional optimisation problem. The same is true if inertial forces are relevant.
- The method would be difficult to generalise to three dimensions. Although the resulting remeshing issues can be resolved [9,10], the optimisation problem for an arbitrary crack surface would require a suitable parametrisation of the surface and result in a high-dimensional optimisation problem with probably prohibitive CPU time costs.

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